

Can Yesterday’s State Spare the Decoder from Recalibration? DriftGate: Minimal Calibration for Neural Drift in Brain-to-Text BCI

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Abstract

Daily BCI calibration should not be automatic. Prompted calibration trials cost user time, require target-day labels, and may update decoder-facing parameters even when a compatible state already exists. We introduce *DriftGate*, an evidence-triggered minimal calibration policy for long-term brain-to-text BCI deployment. In the public T15 day-conditioned GRU baseline [Card et al., 2024], each session has a learned input layer feeding a shared frozen GRU backbone and phoneme head. We treat these learned input layers as stored *input states* and ask a deployment question: should a new session reuse the previous state, retrieve an older state using unlabeled neural geometry, or escalate to labeled calibration?

On 41 T15 sessions, fixed non-native input states degrade phoneme error rate (PER) from 9.06% native-day decoding to 22.67–34.43%. Moment correction and $K = 20$ input-layer calibration recover only part of this loss. In contrast, reusing the previous input state reaches 12.21% PER with no target-day labels, no gradient updates, and no changes to the frozen decoder—substantially better than $K = 20$ input-layer calibration from a mismatched fixed source. The resulting policy is simple: reuse recent states when possible, use geometry as a risk sensor for stale or unusual sessions, and recalibrate only when the stored-state library appears unreliable. Geometry predicts decoder error and retrieves plausible alternatives, but paired bootstrap and random-window checks show that it is not reliable enough to override recency automatically. Thus *DriftGate* turns BCI recalibration from a calendar event into an evidence-triggered decision.

1 Introduction

Every calibration prompt is a communication opportunity lost. Intracortical brain-computer interfaces (BCIs) can restore communication by mapping neural activity to intended speech or text, and recent speech neuroprostheses have

reached high accuracy. Yet long-term use remains fragile because neural recordings drift across sessions. The usual response is to collect prompted target-day trials and recalibrate. For an assistive communication system, however, the question should not be whether a new calendar day has begun. The question should be whether the decoder actually needs new labels or new parameters.

We ask that question directly: *do we need to recalibrate today?* More specifically, we ask whether a brain-to-text BCI can first try to reuse an already learned decoder-facing state, and only request labeled calibration when there is evidence that stored states are unreliable. This shifts recalibration from a daily ritual to a deployment decision.

We study this decision in the public T15 brain-to-text GRU baseline from a rapidly calibrating speech neuroprosthesis study [Card et al., 2024]. The decoder contains learned day-specific input layers, a shared GRU backbone, and a phoneme output head. Each input layer maps one day’s neural features into the representation expected by the shared recurrent decoder; we call this learned input layer an *input state*. Native-day decoding uses the target session’s own input state, whereas cross-day decoding reuses another stored input state while leaving the GRU and head frozen.

This architecture exposes a missing control layer in long-term BCI deployment. Instead of asking only how to recalibrate faster, we ask when recalibration should be avoided altogether. A new session can reuse the previous input state, retrieve an older candidate using a short unlabeled neural window, or escalate to labeled input-layer calibration when stored states appear unreliable. We call this policy *DriftGate*.

Our contributions are:

- **A deployment frame:** we formulate cross-session brain-to-text drift as *input-state triage*: reuse, retrieve, or recalibrate while keeping the shared GRU and phoneme head frozen.
- **A surprising zero-label result:** previous-state reuse reaches 12.21% PER in the $K = 20$ beginning-of-day setting, closing about 75% of the fixed-middle-to-native gap with no target-day labels, no gradient updates, and no decoder changes.
- **A burden-aware comparison:** from a mismatched fixed source, $K = 20$ input-layer calibration reaches only 18.38% PER; in this setting, choosing yesterday’s

Recency-aware adaptation ladder for cross-session BCI drift

Estimate new-session geometry → retrieve a compatible input state → decode with the GRU frozen

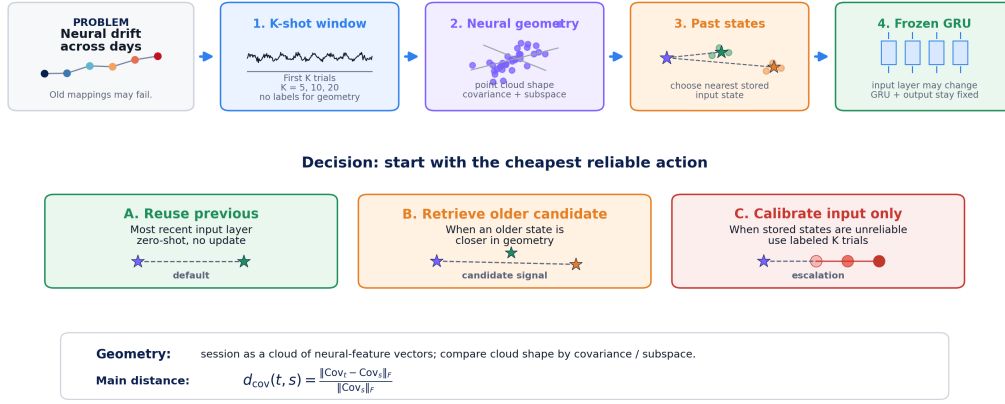


Figure 1: **DriftGate: evidence-triggered recalibration for cross-session BCI drift.** A short beginning-of-day neural window estimates target geometry. The system then chooses the cheapest reliable action: reuse the previous input state, retrieve an older compatible candidate, or escalate to labeled input-layer calibration, while keeping the GRU backbone and phoneme head frozen.

state beats training on 20 labeled calibration trials.

- **A guardrail for recency:** recent previous states are near-native, stale previous states degrade sharply, and geometry predicts decoder error but is not reliable enough to override recency automatically.
- **A replication of the ladder principle:** in T12, fixed old reuse is highly degraded, while previous-day reuse recovers much of the lost accuracy without introducing a new adaptation mechanism.

The claim is intentionally deployment-focused. We do not argue that stored-state reuse replaces supervised retraining with abundant labels. We argue that a BCI should not ask the user for calibration data until a cheaper action has failed: first try a recent compatible state, inspect stale states with geometry, and reserve labeled calibration for evidence of unreliability.

2 Related Work

Brain-to-text BCIs have progressed from handwriting-based decoding to high-performance speech neuroprostheses, but long-term use still requires session-specific adaptation or recalibration as neural recordings change over time [Willett et al., 2021; Willett et al., 2023; Card et al., 2024]. Cross-session stability has also been studied through neural geometry, manifold alignment, continual recalibration, and session-invariant learning [Degenhart et al., 2020; Fan et al., 2023; Karpowicz et al., 2025; Zhang et al., 2026]. Our setting is narrower and more operational: rather than learning a new invariant decoder, we ask whether a frozen decoder with a library of learned input states can decide when to skip, delay, or escalate recalibration. This decision is burden-aware: it compares accuracy to the need for target-day labels, gradient updates, and changed parameters.

3 Data and Decoder

T15 main analysis. The main experiments use 41 T15 validation sessions from the public brain-to-text copy-task dataset associated with Card et al. [Card et al., 2024]. Neural features are passed through the public T15 day-conditioned GRU checkpoint and decoded greedily at the phoneme level after standard CTC collapse. Lower phoneme error rate (PER) is better.

Decoder and input states. The public decoder contains learned day-specific input layers, a shared GRU backbone, and an output layer over phoneme classes. Each input layer projects one day’s neural features into the shared GRU representation. We call the stored input-layer weights for a session an *input state*. Native-day decoding uses the target session’s state; cross-day tests use a fixed or selected non-native source state while keeping the GRU and output head unchanged.

Beginning-of-day setting. We use the first $K \in \{5, 10, 20\}$ target trials for geometry estimation or supervised input calibration, and evaluate only on the remaining trials. For policies that require a past source, the first chronological session is excluded and native/fixed baselines are recomputed on the same remaining-trial evaluation sets. T15 is the main decoder-facing dataset. We additionally use T12 with a public Brain-to-Text Benchmark 2024 checkpoint as a replication of the source-reuse pattern. T16/T17 lack the sentence-level targets, phoneme labels, train/test sentence splits, and reusable decoder checkpoints needed for the same analysis, so they are treated only as supporting cue/behavior checks.

4 DriftGate: Evidence-Triggered Recalibration

4.1 Metrics and Actions

For each validation trial, we compute greedy phoneme predictions from the CTC output stream and apply CTC collapse.

For trials $\mathcal{D}$,

$$\text{PER}(\mathcal{D}) = \frac{\sum_{i \in \mathcal{D}} \text{edit}(y_i, \hat{y}_i)}{\sum_{i \in \mathcal{D}} |y_i|}.$$

Table-level PER values are phoneme-weighted aggregates over evaluation trials; per-session counts are used only for win/loss/tie statements.

DriftGate evaluates policies by both decoding accuracy and calibration burden. A policy may use no target-day signal, unlabeled target-day neural activity, or labeled target-day calibration trials; it may leave parameters unchanged, update only an input transform, or require heavier decoder retraining. The ladder is deliberately conservative:

Start with the cheapest reliable action. Reuse the previous input state by default; use unlabeled geometry to detect risk and nominate older candidates; request labeled calibration only when stored states appear unreliable.

This policy does not assume that recency always wins. It uses recency as the default because nearby sessions are often compatible, and uses geometry as a smoke detector for stale or unusual sessions.

4.2 Adaptation and Source Selection

We compare native-day decoding against three fixed non-native source layers: early (t15.2023.08.13), middle (t15.2023.11.26), and late (t15.2025.04.13). Using the middle source layer, we test no correction, unsupervised mean/scale moment matching, supervised diagonal affine calibration, and supervised input-layer calibration. Learned adapters use the first K labeled trials and are evaluated on the remaining trials. The GRU backbone and output head remain frozen; only the input transform is learned using CTC loss.

Neural feature geometry. For each session, we collapse trials and time bins and treat each time bin as one neural-feature vector in channel space. The covariance matrix summarizes the population-activity cloud. For K -shot source selection, the first K unlabeled target trials estimate target covariance. No target text labels are used for geometry. We use relative Frobenius covariance distance:

$$d_{\text{cov}}(t, s) = \frac{\|\text{Cov}(S_t^{(K)}) - \text{Cov}(S_s)\|_F}{\|\text{Cov}(S_s)\|_F + \epsilon},$$

with $\epsilon = 10^{-8}$. We select the closest past source,

$$s^*(t) = \arg \min_{s \in S_{\text{past}}(t)} d_{\text{cov}}(t, s),$$

and compare it to previous-session reuse, $s_{\text{prev}}(t)$. For diagnostics, we also compute PCA subspace distances using the top 20 principal components and mean principal angle.

4.3 Uncertainty and Robustness Checks

We report paired-bootstrap confidence intervals for the deployment-relevant comparisons. Each bootstrap draw resamples target sessions with replacement and recomputes the paired PER difference; intervals are empirical 2.5th and

97.5th percentiles. These intervals are used to test whether the large source-reuse gains and the small geometry-override effects are stable across sessions.

We also test whether beginning-of-day results depend on using the first K trials. For each eligible T15 target session and each K , we sample 50 random contiguous K -trial windows, estimate geometry from the sampled window, and evaluate decoding after excluding that window. No adapter training is performed in this robustness check. The goal is not to tune a policy on random windows, but to verify that the qualitative source-selection ordering is not an artifact of a particular prefix.

5 Experiments and Results

We first show that the wrong input state can break an otherwise fixed decoder. We then show that drift has measurable geometry, that previous-state reuse is a strong zero-label default, and that geometry should be used as a risk sensor rather than an autopilot.

5.1 The Wrong Input State Breaks the Frozen Decoder

Native-day decoding reaches 9.06% phoneme-weighted PER. When all sessions are forced through non-native source input layers, PER rises to 22.67–34.43%, harming 40/41 sessions for each source choice. The fixed early, middle, and late source states reach 28.81%, 22.67%, and 34.43% PER, respectively. Moment matching remains close to the fixed-source baseline at 21.82%, showing that cross-session mismatch is not merely a mean/variance shift.

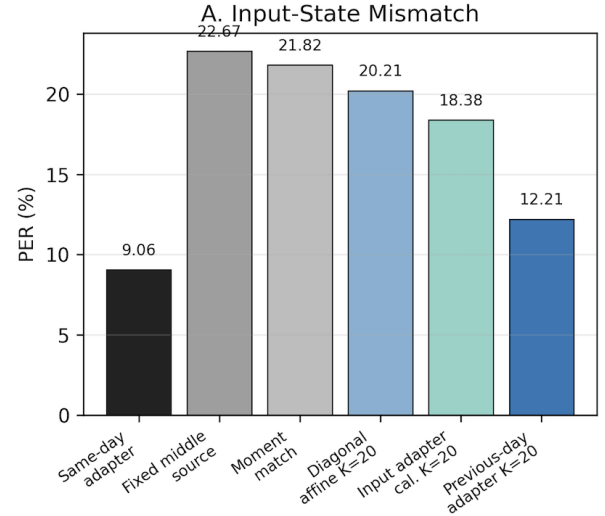


Figure 2: **Input-state mismatch breaks the frozen decoder.** Native-day decoding is near 9% PER, whereas mismatched non-native states and simple moment matching remain far worse.

Supervised diagonal affine and input-layer calibration help, reaching 20.21% and 18.38% PER in the fixed-middle stress setting, but they remain far from 9.06% native decoding. Small supervised calibration is useful, but it does not replace a well-matched prior input state.

Table 1: **T15 decoder-facing stress test.** Fixed-source and moment-correction rows use the middle source t15.2023.11.26. Learned adapter rows use $K = 20$ calibration trials and are evaluated on remaining trials.

Condition	Aggregate PER ↓	Best session	Worst session
Native-day	9.06%	0.80% (11-19)	23.82% (03-14)
Fixed early	28.81%	7.38% (08-13)	58.28% (03-14)
Fixed middle	22.67%	1.20% (11-19)	56.44% (03-30)
Fixed late	34.43%	16.40% (01-12)	49.04% (07-19)
Moment match	21.82%	1.20% (11-19)	56.90% (03-30)
Diag. affine, $K = 20$	20.21%	1.94% (11-26)	53.26% (03-30)
Input layer, $K = 20$	18.38%	2.46% (11-26)	52.23% (03-30)

5.2 Drift Is Structured, Not Random

T15 sessions have temporally structured input geometry. Figure 3 shows that pairwise covariance shift increases with source–target time gap: nearby sessions form a diagonal band of similar geometry, supporting recency as a strong prior.

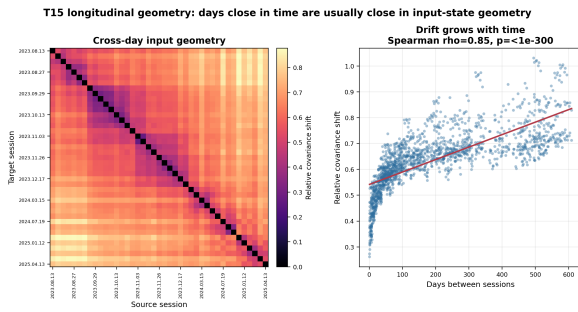


Figure 3: **T15 longitudinal geometry.** Covariance shift forms a near-diagonal structure and increases with source–target time gap (Spearman $\rho = 0.85$).

The full source–target drift atlas shows that geometry is also decoder-relevant. Across all past source–target pairs, source–target time gap and neural geometry predict heldout PER. In T15, time-gap-vs-PER and geometry-vs-PER reach Spearman correlations of $\rho = 0.850$ and $\rho = 0.785$; T12 shows the same pattern with $\rho = 0.813$ and $\rho = 0.538$. Thus cross-session degradation is structured and measurable, not random.

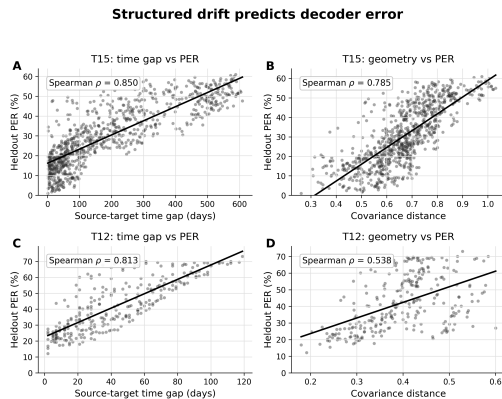


Figure 4: **Structured drift predicts decoder error.** Across source–target pairs, decoder error increases with time gap and geometry distance in both T15 and T12.

5.3 Yesterday’s State Is a Strong Zero-Label Default

In the beginning-of-day setting, the first K trials estimate geometry or train a small adapter, and evaluation is performed only on remaining trials. Past-only geometry selection reaches 14.00%, 13.19%, and 12.74% PER at $K = 5, 10, 20$, far below fixed-middle decoding. However, the immediately previous input layer is even stronger: 13.09%, 12.84%, and 12.21% PER. This is the key result: before asking the user for labels, the system should first ask whether yesterday’s state still works.

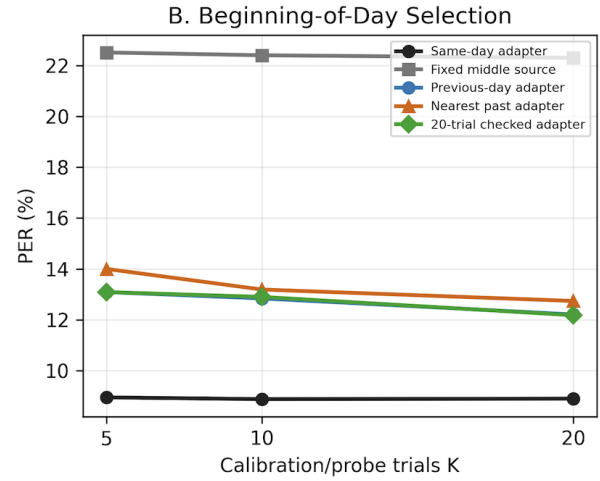


Figure 5: **Beginning-of-day source selection.** Previous-state reuse is a strong zero-label default across $K = 5, 10, 20$, while nearest-past geometry remains close but does not dominate it.

At $K = 20$, previous-state reuse closes approximately 75% of the fixed-middle-to-native gap: fixed-middle decoding on the same remaining trials is 22.30% PER, native-day decoding is 8.90%, and previous-state reuse is 12.21%. Most avoidable error from a wrong fixed source is removed without target-day labels or parameter updates.

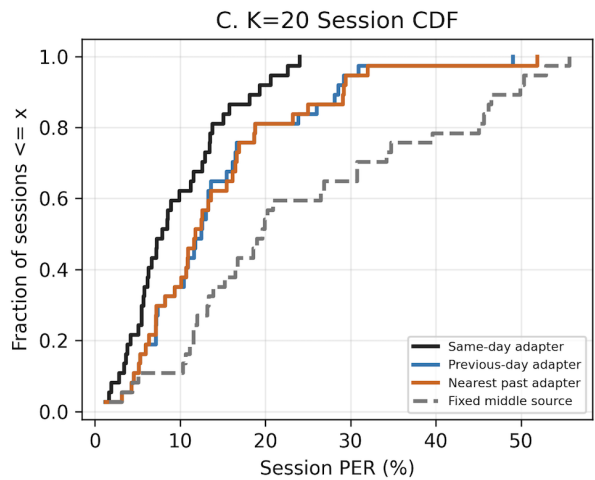


Figure 6: **$K = 20$ session-level distribution.** Previous-state reuse shifts sessions toward lower PER relative to the fixed-middle source, showing that the gain is not only an aggregate effect.

The session-level CDF in Figure 6 shows the same pattern beyond aggregate PER: fixed-middle reuse produces substantially more high-error sessions, while previous-state reuse shifts the distribution toward lower PER without labels or updates.

Table 2: **Beginning-of-day source selection.** Evaluation uses only remaining validation trials after the K -trial prefix. Lower PER is better.

Method	$K = 5$	$K = 10$	$K = 20$
Native-day	8.95%	8.88%	8.90%
Fixed middle	22.51%	22.40%	22.30%
Previous source	13.09%	12.84%	12.21%
K-shot geometry source	14.00%	13.19%	12.74%
Best simple override	13.09%	12.90%	12.18%

5.4 Zero-Label Reuse Beats Small Labeled Calibration

The most deployment-relevant comparison is not just accuracy, but accuracy per unit burden. Table 3 shows that with $K = 5$ or $K = 10$ labeled trials, supervised input-side corrections remain near 20–22% PER, while previous-state reuse stays near 13% PER with no labels. Even at $K = 20$, input-layer calibration from a poor fixed source remains substantially worse than simply reusing the previous state. Small labeled calibration is therefore best treated as an escalation for unreliable stored states, not as the first action on every day.

Table 3: **T15 label-budget curve.** Supervised rows start from the fixed-middle source and update only an input-side transform; previous-state reuse uses no target-day labels or updates.

Method	$K = 5$	$K = 10$	$K = 20$
Previous-state reuse	13.09%	12.84%	12.21%
Diag. affine calibration	21.99%	21.81%	20.21%
Input-layer calibration	21.10%	20.30%	18.38%

Table 4: **The deployment surprise: the cheapest first action is often the strongest.** T15 PER values use the $K = 20$ beginning-of-day setting where applicable.

Method	Target-day signal	Labels / update	PER ↓
Native-day input state	target-day trained state	yes / yes	8.90%
Fixed middle source	none	no / no	22.30%
Previous-state reuse	stored previous state	no / no	12.21%
Geometry retrieval	$K = 20$ unlabeled trials	no / no	12.74%
Best simple override	$K = 20$ unlabeled trials	no / no	12.18%
Input-layer calibration	$K = 20$ labeled trials	yes / yes	18.38%

Table 4 makes the burden tradeoff explicit. Previous-state reuse needs no target-day labels or gradient update; geometry retrieval uses only $K = 20$ unlabeled neural trials. For scale, Card et al. [Card et al., 2024] report a rapid-calibration protocol in which 20 prompted copy-task sentences took approximately 7.5 minutes and provided training labels. Our point is not that labels are never useful. It is that a deployed system can first ask whether a stored state is reliable enough to avoid those labeled updates.

5.5 The Recalibration Ladder: Green, Amber, Red

The operational policy is conservative and easy to interpret. **Green:** if the previous state is recent, reuse it and skip labeled calibration. **Amber:** if the previous state is stale or geometrically unusual, inspect stored candidates with unlabeled geometry. **Red:** if no stored state appears reliable, collect labeled calibration data or use heavier retraining.

Table 5: **Observed T15 triage regimes.** Recalibration becomes an evidence-triggered escalation rather than a daily default.

Condition	Evidence	Triage action	Interpretation
Recent previous state	9.97% PER for 0–3 day gap	Reuse previous	Often skip labeled calibration
Generic $K = 20$ day start	Previous: 12.21%; geometry: 12.74%	Use recency default	No update before risk evidence
Stale previous state	24.54% PER for >14 day gap	Retrieve or calibrate	Recency is not safe
Wrong fixed state	Fixed: 22.67%; input cal.: 18.38%	Calibrate if no reliable state	Labels help but do not replace state choice

This ladder is the central deployment claim. DriftGate does not say “never recalibrate.” It says: do not recalibrate by ritual. Reuse recent states when they are likely reliable, use geometry to detect risk in stale states, and escalate to labeled calibration only when the stored-state library is unreliable.

5.6 Geometry Is a Risk Sensor, Not an Autopilot

The previous-session input layer is the strongest simple baseline, but it is not always the geometrically nearest source. At $K = 20$, geometry selects the previous source in 26 sessions and an older non-previous source in 11 sessions. Among those 11 older selections, the older source wins in 3 cases, previous remains better in 7, and 1 case is tied. When geometry incorrectly selects an older source, the mean penalty relative to previous-session reuse is 3.23 PER points. Thus geometry identifies plausible candidates, but covariance distance alone is not sufficient as a deployment policy.

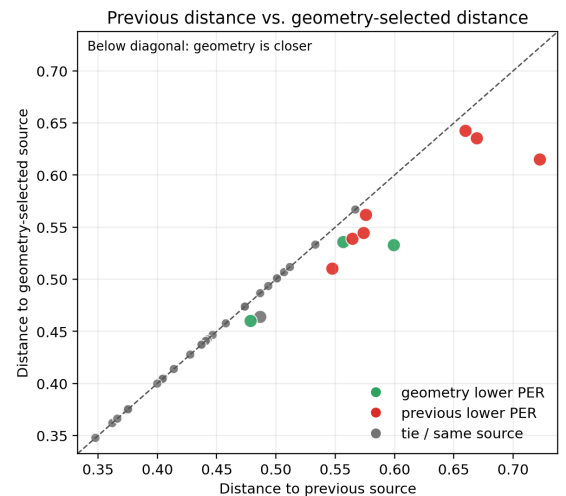


Figure 7: **Geometry is useful but not an autopilot.** Geometry can select a source closer in covariance distance than the previous state, but closer geometry does not always imply lower PER.

Table 6: **Paired-bootstrap uncertainty for deployment-relevant comparisons.** Deltas are PER percentage points for the second policy minus the first policy; positive values mean the second policy is worse.

Comparison	Δ PER pp	95% CI
T15 previous vs fixed middle, $K = 20$	+10.09	[6.56, 13.97]
T15 previous vs same-day, $K = 20$	-4.74	[-6.70, -3.04]
T15 previous vs nearest past, $K = 20$	+0.25	[-0.15, 0.75]
T15 fixed middle vs input cal., $K = 20$	-2.89	[-4.15, -1.77]
T12 previous vs fixed old	+42.54	[36.88, 46.85]

The bootstrap results in Table 6 separate robust effects from marginal ones. Previous-state reuse strongly improves over fixed-middle decoding in T15 and over a fixed old source in T12. Same-day input states still define a lower-error upper reference, and input-layer calibration from a fixed source helps, but its gain is much smaller than choosing a compatible stored state. In contrast, automatic nearest-past geometry selection is not reliably better than previous-state reuse: its paired interval crosses zero. This reinforces the decision to use geometry as a candidate and risk signal rather than as an automatic override.

Random-window robustness yields the same qualitative ordering. Across 50 random contiguous windows per eligible T15 session, previous-state reuse reaches mean PERs of 14.10%, 14.04%, and 14.74% for $K = 5, 10, 20$; nearest-past geometry reaches 14.63%, 14.43%, and 15.04%. Geometry selects a non-previous source in 28.8–31.6% of windows, so it does retrieve alternative candidates, but those alternatives do not improve aggregate decoding. A simple distance-ratio gate likewise does not solve the override problem: at $K = 20$, the best simple override improves only from 12.21% to 12.18%, and a leave-one-session-out decision stump improves only to 12.15% while performing worse at smaller K . Winning older sources reduce mean principal angle relative to previous by 2.91° , whereas losing older sources do not. We therefore treat geometry as a risk and retrieval signal, not as an automatic override of recency.

5.7 T12 Public-Checkpoint Replication

Using a public Brain-to-Text Benchmark 2024 checkpoint, we evaluate T12 in the same spirit: keep the decoder fixed and compare same-day decoding, old-source reuse, previous-day reuse, and geometry-selected reuse. This replication is intentionally limited to source choice so that it tests the same input-state triage question as T15.

Table 7: **T12 public-checkpoint replication of source reuse.** Lower PER is better.

Method	Aggregate PER ↓
Same-day / native	17.87%
Fixed old source	66.22%
Previous-day source	23.00%
Geometry-selected source	23.61%

The T12 pattern supports the same ladder principle without introducing an additional adaptation mechanism. Fixed old reuse is highly degraded, previous-day reuse recovers much of the gap, and geometry-selected reuse remains close to but

slightly worse than the previous-day default. The T12 bootstrap comparison in Table 6 confirms that the previous-day gain over a fixed old source is large and stable across target sessions.

6 Discussion

The surprising result is that the best first response to drift is not necessarily adaptation. In this decoder architecture, choosing a compatible stored input state can outperform small labeled calibration while requiring no target-day labels, no gradient updates, and no parameter changes.

Cross-session adaptation should therefore be framed as triage rather than as a single fixed correction. Native T15 input layers are highly effective, fixed non-native input layers cause large decoder-facing degradation, moment alignment is too weak, and learned input calibration improves PER only partially. The most useful low-cost intervention is often not a new adapter, but a better choice of an existing input state.

The simplicity of previous-state reuse is the point. A long-term BCI should not start every day by collecting labels and changing parameters. It should first audit whether an already learned decoder-facing state remains compatible. If the previous state is recent, reuse it without calibration: previous-state reuse reaches 9.97% PER at 0–3 days. If the previous state is stale, geometry should nominate stored candidates rather than automatically overriding recency: beyond 14 days, previous reuse degrades to 24.54% PER. If no stored state appears reliable and prompted target-day trials are available, residual input calibration is an escalation step; in the fixed-middle stress setting, $K = 20$ calibration improves PER from 22.67% to 18.38%, but remains far from native-day decoding.

6.1 Operational Interpretation

The deployment lesson is that calibration should be treated as a decision under uncertainty, not as a daily default. A new session can fall into three practical regimes. In the green regime, the previous input state is recent and geometrically compatible, so the system can reuse it without labels or parameter updates. In the amber regime, the previous state is stale or geometrically unusual, so the system should inspect older stored states or collect a short additional probe before changing anything. In the red regime, no stored state appears reliable, and labeled calibration becomes justified.

This conservative interpretation is important because geometry is predictive but not decisive. The source–target covariance distance is useful as a label-free warning signal, but it should not silently override recency. In our T15 results, nearest-past geometry retrieves plausible alternatives, yet it does not reliably beat previous-state reuse across sessions. The right conclusion is therefore not “always choose the nearest geometry source,” but rather “use geometry to decide when memory may be unsafe.” This is why DriftGate uses geometry as a risk sensor and retrieval signal, not as an autopilot.

The state-library view also explains why small labeled calibration can lose to previous-state reuse. A recent input state already contains a learned decoder-facing map that may remain compatible with the target day. By contrast, a $K = 20$

input-layer update from a mismatched fixed source must re-construct a usable input map from very little labeled data. Labels are still valuable, but they should be spent when they are likely to change the decision. In this sense, DriftGate complements future adaptation methods: stronger fine-tuning, language-model decoding, or richer uncertainty estimates can improve the red-regime response, while the ladder decides whether that response is needed today.

This burden view sharpens the deployment claim. Source reuse is not meant to dominate fully supervised retraining with abundant labels; it is a pre-retraining action that can reduce unnecessary calibration days. Additional all-source analyses suggest that source choice alone is close to saturated: in T15, a deployable geometry-plus- $K = 20$ calibration probe improves previous reuse only from 12.21% to 12.04% PER, while oracle best-source choice reaches 11.81%. Further gains likely require target-day correction rather than better source choice alone.

This framing complements stronger decoder-backbone approaches. Pretrained speech models or language-model decoding may improve absolute performance, while DriftGate addresses a different deployment question: how to keep a fixed decoder usable across sessions with minimal calibration. The core principle is general: before training, test whether memory is enough.

7 Limitations

This study is a retrospective decoder-facing deployment-policy study, not a new decoding architecture or a clinical deployment study. The main evidence comes from T15, with T12 serving as a public-checkpoint replication; T16/T17 are only cue/behavior support because they lack sentence-level targets, phoneme labels, train/test sentence splits, and reusable decoder checkpoints. We report greedy PER rather than full WER or language-model decoding.

The triage rule identifies empirical regimes in which calibration appears avoidable or necessary, but it is not yet a prospectively validated clinical gate. We also do not benchmark wall-clock training time for full-network fine-tuning or from-scratch retraining; those require separate matched compute, data, and stopping-rule experiments. Finally, covariance geometry can nominate candidates and detect risk, but richer confidence features, memory management, prospective validation, and safety constraints are needed before geometry can safely override recency in deployment.

8 Conclusion

Daily BCI calibration should not be automatic. In the T15 public GRU baseline, fixed non-native input layers break the decoder, moment correction is too weak, and learned input adaptation improves PER only partially. The largest practical improvement comes from selecting a compatible past input layer. In the $K = 20$ beginning-of-day setting, previous-state reuse reaches 12.21% PER with no target-day labels, no gradient update, and no change to the frozen GRU backbone or output head, closing about 75% of the fixed-source-to-native gap and outperforming $K = 20$ input-layer calibration from a mismatched fixed source.

The answer is not to avoid calibration forever; it is to trigger calibration by evidence. Recent previous states can often be reused, stale states should be inspected by geometry or escalated, and labeled calibration should be reserved for sessions whose stored states are unreliable. Across T15 and T12, source–target time gap and neural geometry predict decoder error, supporting a recency-aware, geometry-guided input-state triage system. DriftGate turns recalibration from a calendar event into a decision: reuse when memory is reliable, retrieve when geometry warns, and recalibrate only when evidence demands it.

Ethical Statement

This work uses publicly released intracortical BCI datasets and reports only aggregate analyses. It does not attempt to infer private inner speech, identify participants, or deploy a decoder. Any deployed calibration-skipping policy should include explicit intent gating, conservative safety checks, user override, and appropriate clinical oversight.

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